c h a p t e r **7:** problems and solutions

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2. A raindrop of mass 3.35×10^{-5} kg falls vertically at **W** constant speed under the influence of gravity and air resistance. Model the drop as a particle. As it falls 100 m, what is the work done on the raindrop (a) by the gravitational force and (b) by air resistance?

(a) The work done on the raindrop by the gravitational force is given by

$$W = mgh = \left(3.35 \times 10^{-5}\ \text{kg}\right)\left(9.80\ \text{m/s}^2\right)(100\ \text{m}) = \boxed{3.28 \times 10^{-2}\ \text{J}}$$

(b) Since the raindrop is falling at constant velocity, all forces acting on the drop must be in balance, and $R = mg$, so

$$W_{\text{air resistance}} = \boxed{-3.28 \times 10^{-2}\ \text{J}}$$

3. In 1990, Walter Arfeuille of Belgium lifted a 281.5-kg object through a distance of 17.1 cm using only his teeth. (a) How much work was done on the object by Arfeuille in this lift, assuming the object was lifted at constant speed? (b) What total force was exerted on Arfeuille's teeth during the lift?

(a) The work done by a constant force is given by

$$W = Fd\cos\theta$$

where θ is the angle between the force and the displacement of the object. In this case, $F = -mg$ and $\theta = 180°$, giving

$$W = (281.5\text{ kg})(9.80\text{ m/s}^2)[(17.1\text{ cm})(1\text{ m}/10^2\text{ cm})] = \boxed{472\text{ J}}$$

(b) If the object moved upward at constant speed, the net force acting on it was zero. Therefore, the magnitude of the upward force applied by the lifter must have been equal to the weight of the object:

$$F = mg = (281.5\text{ kg})(9.80\text{ m/s}^2) = 2.76 \times 10^3\text{ N} = \boxed{2.76\text{ kN}}$$

5. A block of mass $m = 2.50$ kg is pushed a distance $d = 2.20$ m along a frictionless, horizontal table by a constant applied force of magnitude $F = 16.0$ N directed at an angle $\theta = 25.0°$ below the horizontal as shown in Figure P7.5. Determine the work done on the block by (a) the applied force, (b) the normal force exerted by the table, (c) the gravitational force, and (d) the net force on the block.

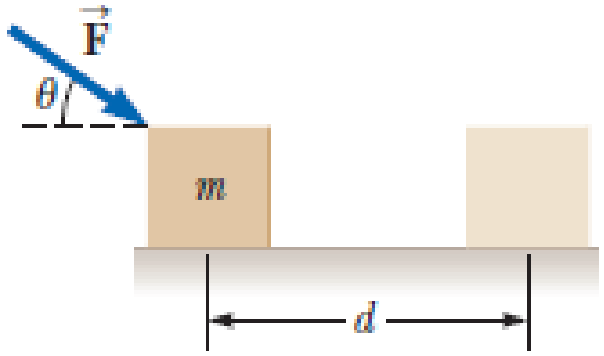


Figure P7.5

you know by thinking about it, without relying on an equation in a list.

The definition of work by a constant force is $W = F\Delta r \cos\theta$.

(a) The applied force does work given by

$$W = F\Delta r \cos\theta = (16.0\text{ N})(2.20\text{ m})\cos 25.0° = \boxed{31.9\text{ J}}$$

(b), (c) The normal force and the weight are both at 90° to the displacement in any time interval. Both do $\boxed{0}$ work.

(d) $\sum W = 31.9\text{ J} + 0 + 0 = \boxed{31.9\text{ J}}$

14. The force acting on a particle varies as shown in Figure P7.14. Find the work done by the force on the particle as it moves (a) from $x = 0$ to $x = 8.00$ m, (b) from $x = 8.00$ m to $x = 10.0$ m, and (c) from $x = 0$ to $x = 10.0$ m.

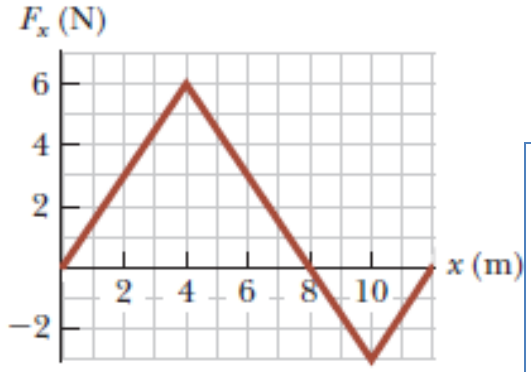
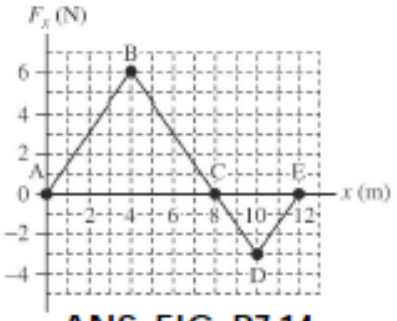


Figure P7.14

$$W = \int_i^f F\,dx = \text{area under curve from } x_i \text{ to } x_f$$



ANS. FIG. P7.14

(a) $x_i = 0$ and $x_f = 800$ m

$W_{0\to 8}$ = area of triangle ABC

$$= \left(\frac{1}{2}\right) AC \times \text{height}$$

$$W_{0\to 8} = \left(\frac{1}{2}\right) \times 8.00\ \text{m} \times 6.00\ \text{N}$$

$$= \boxed{24.0\ \text{J}}$$

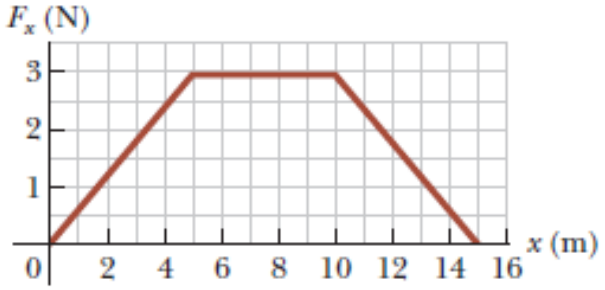
(b) $x_i = 8.00$ m and $x_f = 10.0$ m

$$W_{8\to 10} = \text{area of } \Delta CDE = \left(\frac{1}{2}\right) CE \times \text{height},$$

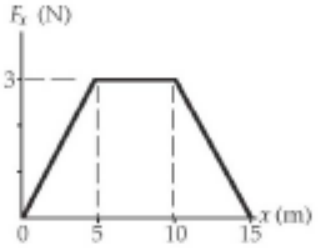
$$W_{8\to 10} = \left(\frac{1}{2}\right) \times (2.00\ \text{m}) \times (-3.00\ \text{N}) = \boxed{-3.00\ \text{J}}$$

(c) $W_{0\to 10} = W_{0\to 8} + W_{0\to 10} = 24.0 + (-3.00) = \boxed{21.0\ \text{J}}$

15. W A particle is subject to a force F_x that varies with position as shown in Figure P7.15. Find the work done by the force on the particle as it moves (a) from $x = 0$ to $x = 5.00$ m, (b) from $x = 5.00$ m to $x = 10.0$ m, and (c) from $x = 10.0$ m to $x = 15.0$ m. (d) What is the total work done by the force over the distance $x = 0$ to $x = 15.0$ m?



We use the graphical representation of the definition of work. W equals the area under the force-displacement curve. This definition is still written $W = \int F_x\,dx$ but it is computed geometrically by identifying triangles and rectangles on the graph.



ANS. FIG. P7.15

(a) For the region $0 \le x \le 5.00$ m,

$$W = \frac{(3.00\text{ N})(5.00\text{ m})}{2} = \boxed{7.50\text{ J}}$$

(b) For the region $5.00 \le x \le 10.0$, $W = (3.00\text{ N})(5.00\text{ m}) = \boxed{15.0\text{ J}}$

(c) For the region $10.00 \le x \le 15.0$, $W = \dfrac{(3.00\text{ N})(5.00\text{ m})}{2} = \boxed{7.50\text{ J}}$

(d) For the region $0 \le x \le 15.0$, $W = (7.50 + 7.50 + 15.0)\text{ J} = \boxed{30.0\text{ J}}$

17. AMT M When a 4.00-kg object is hung vertically on a certain light spring that obeys Hooke's law, the spring stretches 2.50 cm. If the 4.00-kg object is removed, (a) how far will the spring stretch if a 1.50-kg block is hung on it? (b) How much work must an external agent do to stretch the same spring 4.00 cm from its unstretched position?

When the load of mass $M = 4.00$ kg is hanging on the spring in equilibrium, the upward force exerted by the spring on the load is equal in magnitude to the downward force that the Earth exerts on the load, given by $w = Mg$. Then we can write Hooke's law as $Mg = +kx$. The spring constant, force constant, stiffness constant, or Hooke's-law constant of the spring is given by

$$k = \frac{F}{y} = \frac{Mg}{y} = \frac{(4.00\ \text{kg})(9.80\ \text{m/s}^2)}{2.50 \times 10^{-2}\ \text{m}} = 1.57 \times 10^{3}\ \text{N/m}$$

(a) For the 1.50-kg mass,

$$y = \frac{mg}{k} = \frac{(1.50\ \text{kg})(9.80\ \text{m/s}^2)}{1.57 \times 10^{3}\ \text{N/m}} = 0.009\ 38\ \text{m} = \boxed{0.938\ \text{cm}}$$

(b) $$\text{Work} = \frac{1}{2}ky^2 = \frac{1}{2}\left(1.57 \times 10^{3}\ \text{N/m}\right)\left(4.00 \times 10^{-2}\ \text{m}\right)^2 = \boxed{1.25\ \text{J}}$$

20. A light spring with spring constant 1 200 N/m is hung from an elevated support. From its lower end hangs a second light spring, which has spring constant 1 800 N/m. An object of mass 1.50 kg is hung at rest from the lower end of the second spring. (a) Find the total extension distance of the pair of springs. (b) Find the effective spring constant of the pair of springs as a system. We describe these springs as *in series*.

The same force makes both light springs stretch.

(a) The hanging mass moves down by

$$x = x_1 + x_2 = \frac{mg}{k_1} + \frac{mg}{k_2} = mg\left(\frac{1}{k_1} + \frac{1}{k_2}\right)$$

$$= (1.5\text{ kg})(9.8\text{ m/s}^2)\left(\frac{1}{1\,200\text{ N/m}} + \frac{1}{1\,800\text{ N/m}}\right)$$

$$= \boxed{2.04 \times 10^{-2}\text{ m}}$$

(b) We define the effective spring constant as

$$k = \frac{F}{x} = \frac{mg}{mg\left(1/k_1 + 1/k_2\right)} = \left(\frac{1}{k_1} + \frac{1}{k_2}\right)^{-1}$$

$$= \left(\frac{1}{1\,200\text{ N/m}} + \frac{1}{1\,800\text{ N/m}}\right)^{-1} = \boxed{720\text{ N/m}}$$

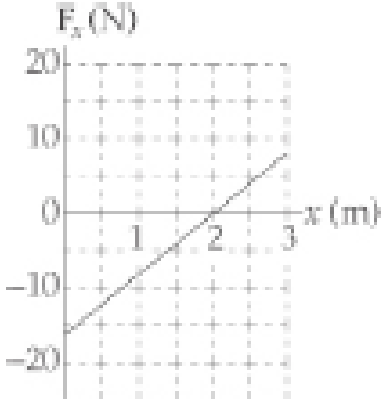
22. Express the units of the force constant of a spring in SI fundamental units.

$$[k] = \left[\frac{F}{x}\right] = \frac{N}{m} = \frac{kg \cdot m/s^2}{m} = \boxed{\frac{kg}{s^2}}$$

26. The force acting on a particle is $F_x = (8x - 16)$, where F is in newtons and x is in meters. (a) Make a plot of this force versus x from $x = 0$ to $x = 3.00$ m. (b) From your graph, find the net work done by this force on the particle as it moves from $x = 0$ to $x = 3.00$ m.

The force is given by $F_x = (8x - 16)$ N.

(a) See ANS. FIG. P7.26 to the right.



(b) $$W_{net} = \frac{-(2.00\text{ m})(16.0\text{ N})}{2} + \frac{(1.00\text{ m})(8.00\text{ N})}{2}$$

$$= \boxed{-12.0\text{ J}}$$

29. A force $\vec{\mathbf{F}} = (4x\hat{\mathbf{i}} + 3y\hat{\mathbf{j}})$, where $\vec{\mathbf{F}}$ is in newtons and **W** x and y are in meters, acts on an object as the object moves in the x direction from the origin to $x = 5.00$ m. Find the work $W = \int \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$ done by the force on the object.

$$W = \int_i^f \vec{F} \cdot d\vec{r} = \int_0^{5\,\mathrm{m}} \left(4x\hat{i} + 3y\hat{j}\right) \mathrm{N} \cdot dx\hat{i}$$

$$\int_0^{5\,\mathrm{m}} (4\ \mathrm{N/m})x\,dx + 0 = (4\ \mathrm{N/m})\frac{x^2}{2}\Bigg|_0^{5\,\mathrm{m}} = \boxed{50.0\ \mathrm{J}}$$

31. A 3.00-kg object has a velocity $(6.00\hat{\mathbf{i}} - 2.00\hat{\mathbf{j}})$ m/s. **W** (a) What is its kinetic energy at this moment? (b) What is the net work done on the object if its velocity changes to $(8.00\hat{\mathbf{i}} + 4.00\hat{\mathbf{j}})$ m/s? (*Note:* From the definition of the dot product, $v^2 = \vec{\mathbf{v}} \cdot \vec{\mathbf{v}}$.)

$$\vec{\mathbf{v}}_i = \left(6.00\hat{\mathbf{i}} - 1.00\hat{\mathbf{j}}\right) \text{ m/s}^2$$

(a) $$v_i = \sqrt{v_{ix}^2 + v_{iy}^2} = \sqrt{37.0} \text{ m/s}$$

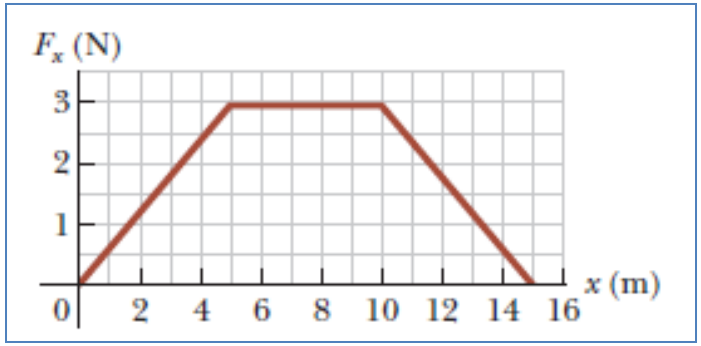
$$K_i = \frac{1}{2}mv_i^2 = \frac{1}{2}(3.00 \text{ kg})\left(37.0 \text{ m}^2/\text{s}^2\right) = \boxed{55.5 \text{ J}}$$

(b) $$\vec{\mathbf{v}}_f = 8.00\hat{\mathbf{i}} + 4.00\hat{\mathbf{j}}$$

$$v_f^2 = \vec{\mathbf{v}}_f \cdot \vec{\mathbf{v}}_f = 64.0 + 16.0 = 80.0 \text{ m}^2/\text{s}^2$$

$$\Delta K = K_f - K_i = \frac{1}{2}m\left(v_f^2 - v_i^2\right) = \frac{3.00}{2}(80.0) - 55.5 = \boxed{64.5 \text{ J}}$$

34. A 4.00-kg particle is subject to a net force that varies **W** with position as shown in Figure P7.15. The particle starts from rest at $x = 0$. What is its speed at (a) $x = 5.00$ m, (b) $x = 10.0$ m, and (c) $x = 15.0$ m?



(a) $\Delta K = K_f - K_i = \frac{1}{2}mv_f^2 - 0 = \sum W =$ (area under curve from $x = 0$ to $x = 5.00$ m)

$$v_f = \sqrt{\frac{2(\text{area})}{m}} = \sqrt{\frac{2(7.50\ \text{J})}{4.00\ \text{kg}}} = \boxed{1.94\ \text{m/s}}$$

(b) $\Delta K = K_f - K_i = \frac{1}{2}mv_f^2 - 0 = \sum W =$ (area under curve from $x = 0$ to $x = 10.0$ m)

$$v_f = \sqrt{\frac{2(\text{area})}{m}} = \sqrt{\frac{2(22.5\ \text{J})}{4.00\ \text{kg}}} = \boxed{3.35\ \text{m/s}}$$

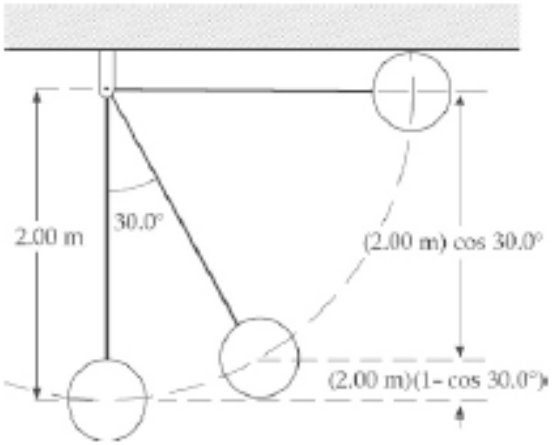
(c) $\Delta K = K_f - K_i = \frac{1}{2}mv_f^2 - 0 = \sum W =$ (area under curve from $x = 0$ to $x = 15.0$ m)

$$v_f = \sqrt{\frac{2(\text{area})}{m}} = \sqrt{\frac{2(30.0\ \text{J})}{4.00\ \text{kg}}} = \boxed{3.87\ \text{m/s}}$$

42. A 400-N child is in a swing that is attached to a pair of ropes 2.00 m long. Find the gravitational potential energy of the child–Earth system relative to the child's lowest position when (a) the ropes are horizontal, (b) the ropes make a 30.0° angle with the vertical, and (c) the child is at the bottom of the circular arc.

(a) We take the zero configuration of system potential energy with the child at the lowest point of the arc. When the swing is held horizontal initially, the initial position is 2.00 m above the zero level. Thus,

$$U_g = mgy$$
$$= (400\ \text{N})(2.00\ \text{m})$$
$$= \boxed{800\ \text{J}}$$



ANS. FIG. P7.42

(b) From the sketch, we see that at an angle of 30.0° the child is at a vertical height of (2.00 m) (1 – cos 30.0°) above the lowest point of the arc. Thus,

$$U_g = mgy = (400\ \text{N})(2.00\ \text{m})(1 - \cos 30.0°) = \boxed{107\ \text{J}}$$

(c) The zero level has been selected at the lowest point of the arc. Therefore, $\boxed{U_g = 0}$ at this location.

43. M A 4.00-kg particle moves from the origin to position ©, having coordinates $x = 5.00$ m and $y = 5.00$ m (Fig. P7.43). One force on the particle is the gravitational force acting in the negative y direction. Using Equation 7.3, calculate the work done by the gravitational force on the particle as it goes from O to © along (a) the purple path, (b) the red path, and (c) the blue path. (d) Your results should all be identical. Why?

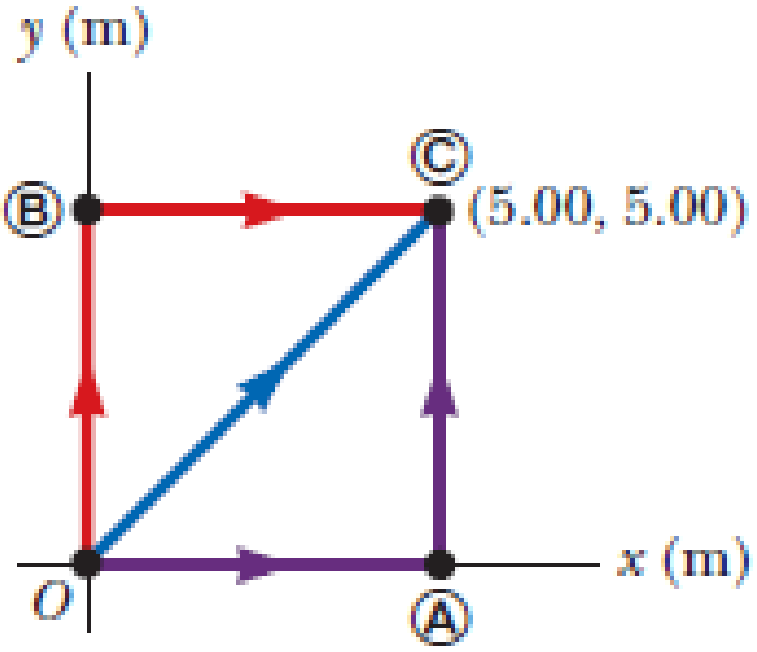
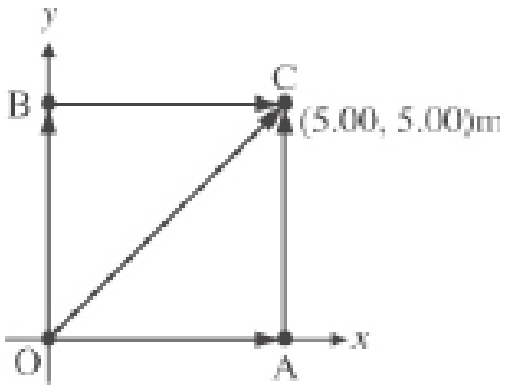


Figure P7.43
Problems 43 through 46.

The gravitational force is downward:

$$F_g = mg = (4.00\ \text{kg})(9.80\ \text{m/s}^2) = 39.2\ \text{N}$$



ANS. FIG. P7.43

(a) Work along OAC = work along OA + work along AC

$$= F_g(\text{OA})\cos 90.0° + F_g(\text{AC})\cos 180°$$

$$= (39.2\ \text{N})(5.00\ \text{m})(0) + (39.2\ \text{N})(5.00\ \text{m})(-1)$$

$$= \boxed{-196\ \text{J}}$$

(b) W along OBC = W along OB + W along BC

$$= (39.2\ \text{N})(5.00\ \text{m})\cos 180° + (39.2\ \text{N})(5.00\ \text{m})\cos 90.0°$$

$$= \boxed{-196\ \text{J}}$$

(c) Work along OC = $F_g(\text{OC})\cos 135°$

$$= (39.2\ \text{N})\left(5.00 \times \sqrt{2}\ \text{m}\right)\left(-\frac{1}{\sqrt{2}}\right) = \boxed{-196\ \text{J}}$$

(d) The results should all be the same, since the gravitational force is conservative.

44. (a) Suppose a constant force acts on an object. The force does not vary with time or with the position or the velocity of the object. Start with the general definition for work done by a force

$$W = \int_i^f \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$$

and show that the force is conservative. (b) As a special case, suppose the force $\vec{\mathbf{F}} = (3\hat{\mathbf{i}} + 4\hat{\mathbf{j}})$ N acts on a particle that moves from *O* to © in Figure P7.43. Calculate the work done by $\vec{\mathbf{F}}$ on the particle as it moves along each one of the three paths shown in the figure and show that the work done along the three paths is identical.

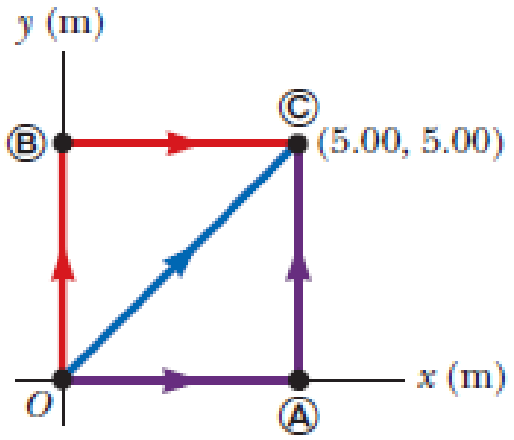


Figure P7.43
Problems 43 through 46.

(a) $W = \int \vec{F} \cdot d\vec{r}$, and if the force is constant, this can be written as

$W = \vec{F} \cdot \int d\vec{r} =$ $\boxed{\vec{F} \cdot \left(\vec{r}_f - \vec{r}_i\right)}$, which depends only on the end points,

$\boxed{\text{and not on the path}}$.

(b) $W = \int \vec{F} \cdot d\vec{r} = \int \left(3\hat{i} + 4\hat{j}\right) \cdot \left(dx\hat{i} + dy\hat{j}\right)$

$= (3.00\ \text{N}) \int_0^{5.00\ \text{m}} dx + (4.00\ \text{N}) \int_0^{5.00\ \text{m}} dy$

$W = (3.00\ \text{N})x\Big|_0^{5.00\ \text{m}} + (4.00\ \text{N})y\Big|_0^{5.00\ \text{m}} = 15.0\ \text{J} + 20.0\ \text{J} = \boxed{35.0\ \text{J}}$

The same calculation applies for all paths.

49. A potential energy function for a system in which a two-dimensional force acts is of the form $U = 3x^3y - 7x$. Find the force that acts at the point (x, y).

$$F_x = -\frac{\partial U}{\partial x} = -\frac{\partial\left(3x^3y - 7x\right)}{\partial x} = -\left(9x^2y - 7\right) = 7 - 9x^2y$$

$$F_y = -\frac{\partial U}{\partial y} = -\frac{\partial\left(3x^3y - 7x\right)}{\partial y} = -\left(3x^3 - 0\right) = -3x^3$$

Thus, the force acting at the point (x, y) is

$$\vec{F} = F_x\hat{i} + F_y\hat{j} = \boxed{\left(7 - 9x^2y\right)\hat{i} - 3x^3\hat{j}}$$

54. The potential energy function for a system of particles is given by $U(x) = -x^3 + 2x^2 + 3x$, where x is the position of one particle in the system. (a) Determine the force F_x on the particle as a function of x. (b) For what values of x is the force equal to zero? (c) Plot $U(x)$ versus x and F_x versus x and indicate points of stable and unstable equilibrium.

(a) $$\vec{F} = -\frac{d}{dx}\left(-x^3 + 2x^2 + 3x\right)\hat{i}$$

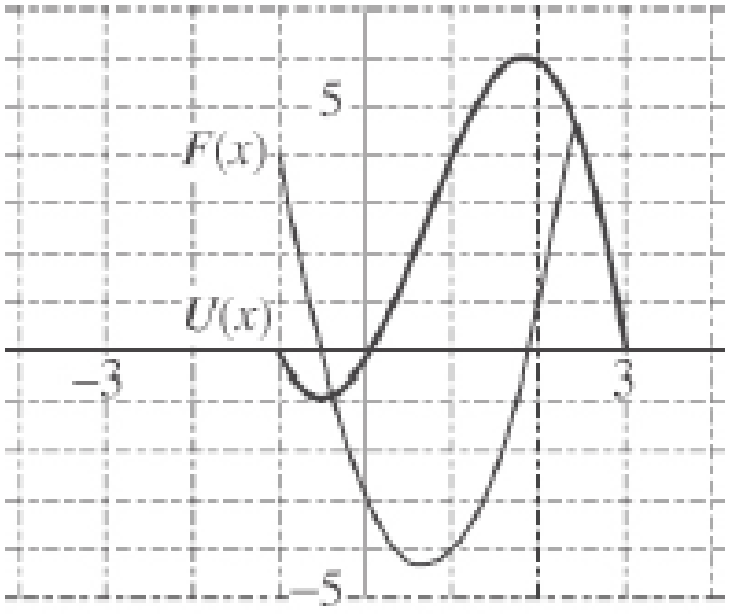
$$= \boxed{\left(3x^2 - 4x - 3\right)\hat{i}}$$

(b) $F = 0$ when

$$x = \boxed{1.87 \text{ and } -0.535}.$$

(c) The stable point is at $x = -0.535$, point of minimum $U(x)$.

The unstable point is at $x = 1.87$, maximum in $U(x)$.



ANS. FIG. P7.54

65. (a) Take $U = 5$ for a system with a particle at position $x = 0$ and calculate the potential energy of the system as a function of the particle position x. The force on the particle is given by $(8e^{-2x})\,\hat{\mathbf{i}}$. (b) Explain whether the force is conservative or nonconservative and how you can tell.

(a) The potential energy of the system at point x is given by 5 plus the negative of the work the force does as a particle feeling the force is carried from $x = 0$ to location x.

$$dU = -F dx$$

$$\int_5^U dU = -\int_0^x 8e^{-2x}\,dx$$

$$U - 5 = -\left(\frac{8}{[-2]}\right)\int_0^x e^{-2x}(-2\,dx)$$

$$U = 5 - \left(\frac{8}{[-2]}\right)e^{-2x}\Big|_0^x = 5 + 4e^{-2x} - 4 \cdot 1 = \boxed{1 + 4e^{-2x}}$$

(b) The force must be conservative because the work the force does on the object on which it acts depends only on the original and final positions of the object, not on the path between them. There is a uniquely defined potential energy for the associated force.